

Preferences

JEB104 Microeconomics I
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Basic Definitions

How do consumers rank their alternative consumption bundles?

Consumption bundles: $x = (x_1, x_2)$, $y = (y_1, y_2)$, $x, y \in X$

Preference relations:

- 1 $x \succeq y$, x is weakly preferred to y , x is at least as good as y
- 2 $x \succ y$, x is strictly preferred to y
- 3 $x \sim y$, x is indifferent to y

Relationship between $\succeq, \succ, \sim$:

If $x \succeq y$ and $y \succeq x \Rightarrow x \sim y$

If $x \succeq y$ and $y \not\succeq x \Rightarrow x \succ y$

Assumptions about Preferences

Rational preferences

- Individual decision maker with rational preferences decides consistently

Rationality assumptions:

① Completeness (complete binary ordering)

Any two bundles from the consumption possibility set can be compared
For any bundles x and y , either $x \succeq y$, or $y \succeq x$, or both (then $x \sim y$).

② Transitivity

If $x \succeq y$ and $y \succeq z$, then $x \succeq z$.

③ Reflexivity

Any bundle is at least as good as itself, $x \succeq x$.

Transitivity of Preferences: Violation



Yellow $\succ$ Black



Red $\succ$ Yellow



Black $\succ$ Red

Transitivity of Preferences: Violation

Suppose Bill needs to choose color of the paint for a room and is always presented with a choice between two shades of green:

C_1	C_2	$C_1 \sim C_2$
C_2	C_3	$C_2 \sim C_3$
C_3	C_4	$C_3 \sim C_4$
C_4	C_5	$C_4 \sim C_5$
C_5	C_6	$C_5 \sim C_6$

By transitivity, it should also hold that $C_1 \sim C_6$. But this may not be the case!

C_1	C_6	$C_1 \succ C_6$
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Transitivity of Preferences: Violation

Intransitivity may also result from combination of multiple transitive preferences

Consider a household formed by Mom (M), Dad (D), and Child (C):

- Need to decide on how they will spend a Friday night
- Options: opera (O), rock concert (R), ice skating show (I)
- Decisions made by majority voting

Their preferences are:

$$O \succ_M R \succ_M I \quad I \succ_D O \succ_D R \quad R \succ_C I \succ_C O$$

As a result, they vote in the following way:

$$O \succ R \quad R \succ I \quad I \succ O$$

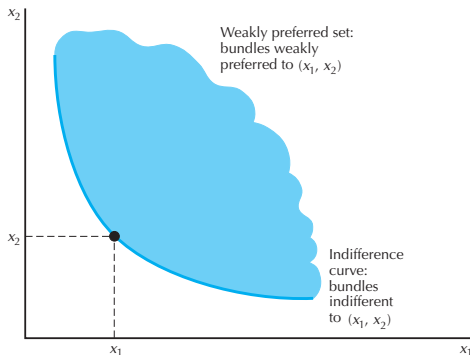
This violates transitivity.

This phenomenon is known Condorcet paradox.

Indifference Curves

Preferences in graphs = indifference curves

- set of bundles among which the consumer is indifferent



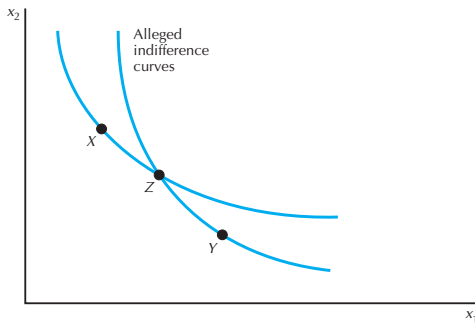
Indifference set of $x = (x_1, x_2)$: $I_x = \{y \in R^2_{+,0} : x \sim y\}$

Weakly preferred set of x : $P_x = \{y \in R^2_{+,0} : y \succeq x\}$

Implications of Rationality

Completeness $\Rightarrow$ every consumption bundle lies on some indifference curve

Transitivity $\Rightarrow$ indifference curves cannot cross



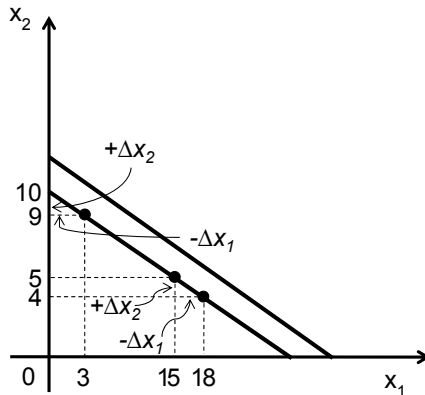
X and Y on different indifference curves $\Rightarrow X \succ Y$, or $Y \succ X$

By transitivity: $X \sim Z \wedge Z \sim Y \Rightarrow X \sim Y \rightarrow$ contradiction

Perfect Substitutes

Consumer willing to substitute one good for the other at a constant rate

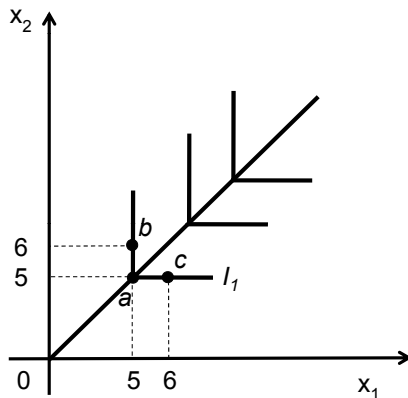
Constant slope



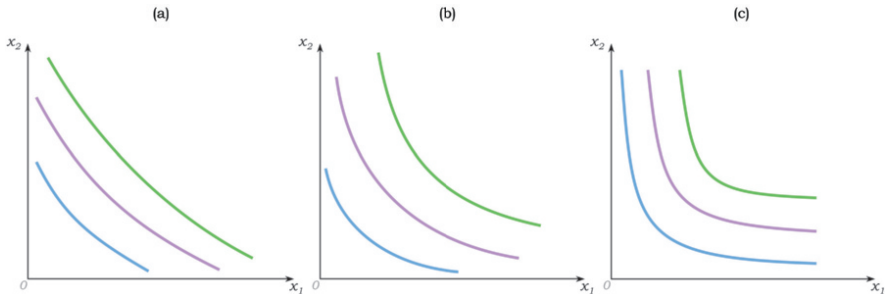
Perfect Complements

Goods are always consumed together in a fixed proportion.

L-shaped indifference curves



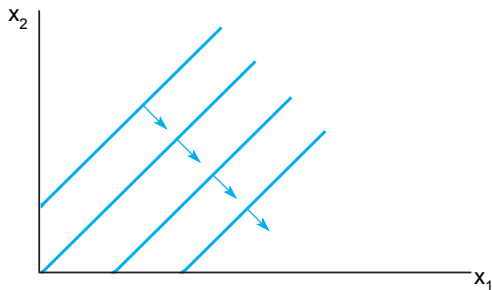
Relative substitutability and complementarity



- Panel (a) – goods are relatively substitutable (close to perfect substitutes case)
- Panel (c) – substitutability is much lower (close to perfect complements case)

Economic Bads

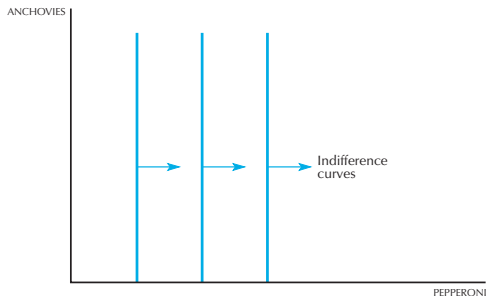
Bad = commodity consumer does not like



Consuming bad x_2 can be compensated by consuming more of x_1 in a given proportion.

Neutral Goods

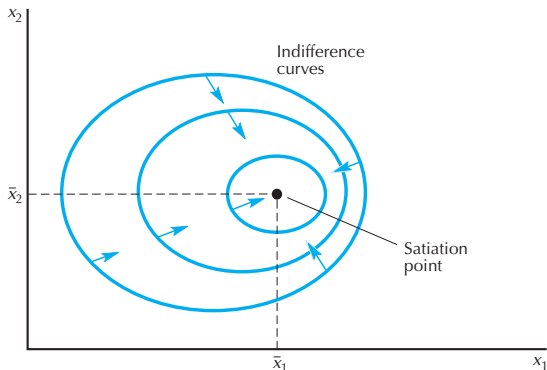
Consumer does not care about the good.



Example: consumer loves pepperoni on a pizza and does not care about how many anchovies he has.

Satiated Preferences

Best bundle $(\bar{x}_1, \bar{x}_2)$ = satiation point = bliss point



Too much of a good $\rightarrow$ economic bad

Well-Behaved Preferences

Indifference curves – various shapes → complicated optimization problem

Additional assumptions imposed on preferences:

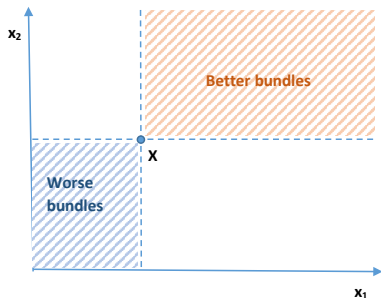
- 1 monotonicity
- 2 continuity
- 3 convexity

Monotonicity

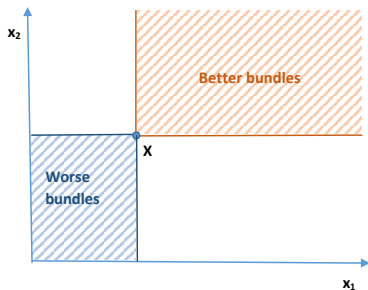
"More is always better, or at least not worse"

(Weakly) monotonic preferences: If $x \gg y \Rightarrow x \succsim y$.

Strictly monotonic preferences: If $x \geq y$ and $x \neq y \Rightarrow x \succ y$.



Monotonicity



Strict monotonicity

Strict monotonicity: indifference curves have negative slope

Continuity

No 'jumps' in people's preferences.

If $x \succ y$, then for z very close to x we have $z \succ y$.

Line connecting any point of the upper contour set (weakly preferred set) and any point of the lower contour set must cross the indifference curve.

Convexity

Averages are preferred to extremes.

People prefer variety.

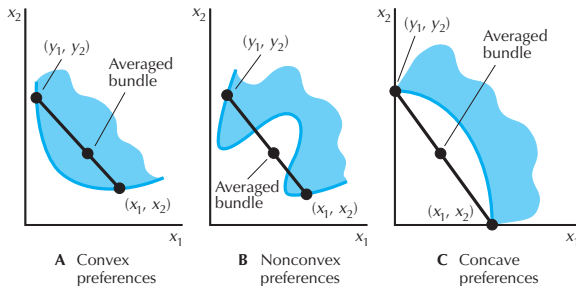
Indifference curve bends toward the origin.

For any $t \in [0, 1]$ and $x, y \in X$, such that $x \sim y$ we have:

$$tx + (1 - t)y \succeq x$$

Weakly preferred set (upper contour set) is a convex set.

Convexity



Panel A: strict convexity (no linear segments)

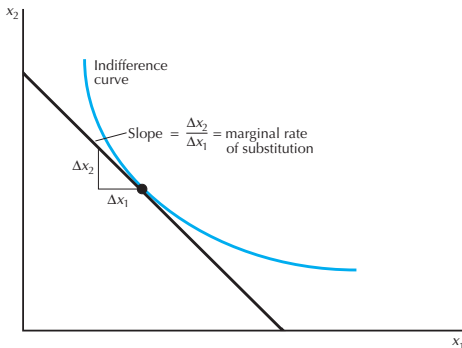
Strict convexity: for any $t \in (0, 1)$ and $x, y \in X$, such that $x \sim y$ we have:

$$tx + (1 - t)y \succ x$$

Marginal Rate of Substitution (MRS)

Rate at which the consumer is just willing to substitute one good for the other while keeping the same utility.

Slope = derivative of the indifference curve at the particular point.



Is MRS negative? Different approaches in different textbooks!!!

Marginal Rate of Substitution

Diminishing MRS (in absolute terms)

- MRS declines as we move along indifference curve down.
- As the consumer is given bundles containing more and more of good 1, she values an individual unit of good 1 less and less.
- Follows from strict convexity.

Specific cases

- Perfect substitutes: constant MRS
- Perfect complements: zero and infinite MRS, not defined

Marginal Rate of Substitution

